Mechanics WS 6

1. a. Consider the function $f(x) = x - \frac{g^2}{x} - 2gln(\frac{x}{g})$, for $x \geq g$.
   - i. Evaluate $f(g)$.
   - ii. Show that $f'(x) = \left(1 - \frac{g}{x}\right)^2$.
   - iii. Explain why $f(x) > 0$ for $x > g$.

   b. A body is moving vertically through a resisting medium, with resistance proportional to its speed. The body is initially fired upwards from the origin with speed v_0. Let y metres be the height of the object above the origin at time t seconds and let g be the constant acceleration due to gravity. Thus $\frac{d^2y}{dt^2} = -(g + kv), where\ k > 0$
   - i. Find v as a function of t, and hence show that $\mathrm{k^2y = (g + kv_0)\left(1 - e^{-kt}\right) - gkt}$
   - ii. Find T, the time taken to reach the maximum height.
   - iii. Show that when $\mathrm{t = 2T, k^2y = (g + kv_0) - \left(\frac{g^2}{g+kv_0}\right) - 2gln\left(\frac{g+kv_0}{g}\right)}$
   - iv. Use this result and part **(a)** to show that the downwards journey takes longer.

2. A projectile is fired with velocity $V = 30m/s$ at an angle of $45°$ to the horizontal. Air resistance is proportional to the velocity. Thus the equations of motion are $\frac{d^2x}{dt^2} = -k\frac{dx}{dt}$ and $\frac{d^2y}{dt^2} = -g - k\frac{dy}{dt}$ where $k = \frac{1}{3}$.
   a. Show that, at time t, the horizontal displacement is $\mathrm{x = 45\left(1 - e^{-\frac{t}{3}}\right)\sqrt{2}}$.
   b. Find a similar expression for the vertical displacement y as a function of t.
   c. Hence show that $\mathrm{y = (1 + \sqrt{2})x - 90log\left(\frac{45\sqrt{2}}{45\sqrt{2}-x}\right)}$
   d. Accurately plot the function in (c) and use it to estimate the range.
   e. By way of comparison, show the trajectory for no air resistance on the same graph.

3. A particle of unit mass is projected vertically upwards from a point O, with initial velocity of u m/s, in a medium whose resistance has magnitude kv^2, where k is a positive constant and v m/s is the velocity after t seconds. Let x be the vertical displacement of the object above the origin after t seconds.

   After reaching its maximum height, the particle then falls back to O, experiencing the same resistance.
   a. Taking upwards as positive, show that $\mathrm{\ddot{x} = -(g + kv^2)}$.
   b. Hence show that the maximum height attained by the particle is $\mathrm{\frac{1}{2k}ln\left(\frac{g+ku^2}{g}\right)}$
   c. Show that the speed of the particle when it returns to O is $\mathrm{\sqrt{\frac{gu^2}{g+ku^2}}}$.
   d. Find the terminal velocity V of any particle of unit mass falling in this medium subject to the resistance $\mathrm{kv^2}$. Hence prove that if the particle in part (i) above is projected with velocity V, it will return to O with speed $\mathrm{\frac{V}{\sqrt{2}}}$.

4. A particle is released from rest $\frac{5\pi}{3}$metres to the left of the origin and travels in a straight line. Its velocity is given by $v^2 = 3 - 6\cos x$.
   a. Find the acceleration and the direction in which the particle first moves. Explain your answer.
   b. Where is the particle stationary again?
   c. Does the particle exhibit SHM?

5. The acceleration of a particle moving in a straight line is given by $\mathrm{\frac{d^2x}{dt^2} = -e^{-2x}}$, where x is the displacement, in metres, from the origin and t is the time elapsed in seconds. Initially the particle moves from the origin with velocity 2 metres per second. Find the velocity of the particle at $\mathrm{x = 1}$, correct to 2 decimal places.

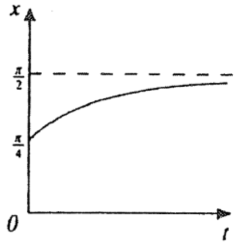
6. A particle initially at rest at $x = 4$, has an acceleration towards the origin given by $\ddot{x} = -4(x + \frac{256}{x^3})$ for $x > 0$

a. Show that the velocity v is given by $v^2 = 4(\frac{256}{x^2} - x^2)$.
b. Explain why the velocity is negative for $0 < x < 4$.
c. Using $\frac{d}{dx}[\cos^{-1}(\frac{x^2}{16})] = \frac{-2x}{\sqrt{256-x^4}}$, find an expression for the time taken to reach position x
d. Find the time for the particle to reach $x = 2\sqrt{2}$

7. A particle moves in a straight line from a fixed point, O, along the x-axis. Its velocity, v m/s is given by: $v^2 = 4x^2 - 9$. Initially the particle is located $\sqrt{3}$ m to the right of O and is moving towards the origin.
   a. Find the set of possible values for x.
   b. Find an expression for the acceleration, a.
   c. Describe the motion of the particle.
8. The displacement x metres of a particle is given by $\mathrm{x} = 7 + 5\sin 3\mathrm{t} + 6\cos 3\mathrm{t}$ where t is the time in seconds
   a. Show that the particle moves in SHM, stating the centre of motion and period T
   b. Find the maximum speed of the particle
   c. Write $5\sin 3\mathrm{t} + 6\cos 3\mathrm{t}$ in the form $\mathrm{R}\cos(3\mathrm{t} - \alpha)$ where $\mathrm{R} > 0$ and $0 < \alpha < 2\pi$
   d. Graph displacement x versus time t of the particle for $0 \le \mathrm{t} \le 2\pi$
9. A particle is moving in a straight line with its acceleration as a function of x given by $\ddot{x} = -e^{-2x}$. It is initially at the origin and is travelling with a velocity of 1 metre per second.
   a. Show that $\ddot{x} = \frac{1}{e^x}$
   b. Hence derive an expression for the displacement of the particle as a function of t
10. The acceleration of a particle is given by $a = x - 1$, where x is the displacement from the origin. Initially it's at the origin moving with a velocity (v) of 1 m/s.
   a. Show that $\mathrm{v}^2 = (\mathrm{x} - 1)^2$
   b. Find the speed of the particle when $\mathrm{x} = \frac{1}{2}$
   c. Show that $\mathrm{x} = 1 - \mathrm{e}^{-\mathrm{t}}$
   d. Will the particle reach $\mathrm{x} = 2$? Justify your answer.
11. The acceleration of a particle P is given by the equation $\ddot{x} = 18x^3 + 27x^2 + 9x$. Initially $x = -2$ and the velocity $v = -6$
   a. Show that $\mathrm{v}^2 = 9\mathrm{x}^2(1 + \mathrm{x})^2$
   b. Hence or otherwise, show that for some constant C, $\int \frac{1}{\mathrm{x}(1+\mathrm{x})}\mathrm{dx} = -3\mathrm{t} + \mathrm{C}$
   c. Find the derivative of $\ln\left(1 + \frac{1}{\mathrm{x}}\right)$
   d. Using the previous results and initial conditions, find x as a function of t
12. A particle is moving in a straight line. At time t seconds it has displacement x metres to the right of a fixed point O on the line and velocity $v\ ms^{-1}$ given by $v = \sin x \cos x$. The particle starts $\frac{\pi}{4}$ metres to the right of O.
   a. Show that $\frac{d}{dx}\ln(\tan x) = \frac{1}{\sin x \cos x}$.
   b. Hence show that the displacement of the particle is given by $x = \tan^{-1}(e^t)$
   c. Find the limiting position of the particle and sketch the graph of x against t.

Answers:

1. (a i) 0, (iii) $f'(x) > 0$ for $x > g$. That is, f is increasing for $x > g$.; (b ii) $T = \frac{1}{k} ln \frac{\mathrm{g}+\mathrm{kV_0}}{g}$
2. (b) $\mathrm{y} = 45\ (2 + \sqrt{2})(1 - \mathrm{e}^{-\frac{\mathrm{t}}{3}})$; (d) $\mathrm{R} \doteqdot 44$, here is the trajectory for $0 \le 2 \le 4$. The dotted line is the trajectory for no air resistances.
3. $V = \sqrt{\frac{g}{k}}$
4. a. $\ddot{x} = 3\sin x$, moves right; b. $x = 2n\pi \pm \frac{\pi}{3}, n \in Z$; c. not SHM
5. $v = 1.57$
6. c. $t = \frac{1}{4}\cos^{-1}\left(\frac{x^2}{16}\right)$; d. $t = \frac{\pi}{12}$
7. a. $x \ge \frac{3}{2}$; b. $a = 4x$; c. from $x = \sqrt{3}$ the particle moves left with speed $\sqrt{3}$. As the acceleration is always directed to right, the particle slows down and stops at $x = \frac{3}{2}$. Due to this acceleration directed to right, the particle changes direction at $x = \frac{3}{2}$ and speeds up continuously
8. $x = 3, T = \frac{2\pi}{3}, 3\sqrt{61}, R = \sqrt{61}, \alpha = \tan^{-1}\frac{5}{6},\ t = 0.08s$
9. b. $x = \ln(t + 1)$
10. b. $\frac{1}{2}$ m/s; d. no (prove graphically)
11. $-\frac{1}{x(x+1)}, x = \frac{2}{e^{3t}-2}$
12. c. limiting position is $\frac{\pi}{2}$ metres to the right of O.


x
π/2
π/4
0
t